Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	University College Dublin, National University of Ireland, Dublin
Short name / acronym used in projects	UCD
Participant Identification Code (PIC), if available	999986384
Website(s)	https://www.ucd.ie/
Country	Ireland
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public bodv ☐ Other
Main address	Belfield, Dublin 4, Ireland
Department(s), institute(s), research group(s) or unit(s) that would be involved	UCD School of Biomolecular and Biomedical Science; UCD Conway Institute of Biomolecular and Biomedical Research
Additional institutional resources relevant to collaboration	Zebrafish Research Facility, UCD Conway Imaging Core, Genomics Core, Proteomics Core, Mass Spectrometry Facility

Main contact person for this profile	Prof. Breandán Kennedy, Professor of Pharmacology
Main contact email	Brendan.kennedy@ucd.ie

Other contact person(s), if relevant	

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Zebrafish models; retinal dystrophies; visual function assays; drug discovery; small molecule screening; neuroprotection; ocular pharmacology; translational validation; angiogenesis; photoreceptors, circulating biomarkers
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
The Kennedy laboratory at UCD focuses on ocular pharmacology and translational visual science, specifically targeting inherited retinal degenerations (IRDs) and acquired eye diseases. Utilizing high-throughput, in vivo zebrafish models of human visual disorders alongside cell-based platforms, the team investigates mechanisms of photoreceptor degeneration, vascular leakage, and visual cycle. A key strength lies in phenotype-based small molecule screening pipelines to identify novel therapeutic candidates, and omics pipelines to identify biomarkers.

Why your team could be relevant to a translational eye-health and vision consortium
Prof. Kennedy's group bridges fundamental disease mechanisms and early-stage drug discovery, providing a crucial in vivo link between in vitro target identification and mammalian/clinical evaluation. The team contributes validated zebrafish disease models for rapid target validation, functional visual testing (e.g., visual behavior/OKR), and pharmacodynamic assessment of novel formulations or delivery systems. This expertise complements IOBA's focus on advanced therapies, neuroprotection, and biomarkers, offering interdisciplinary synergy to accelerate translational ocular therapies

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
High-throughput in vivo small molecule screening using zebrafish models of visual impairment.

Functional visual recovery assays (optokinetic response/OKR, visual behavior analytics). *Ocular angiogenesis, vascular permeability, and neuroprotection drug testing platforms.* *Target validation, histological analysis, bioimaging, and molecular profiling of retinal disease.*

Infrastructure, equipment, facilities or datasets relevant to consortium work
• *Fully equipped, high-capacity Zebrafish Research Facility with automated behavioral tracking systems.* • *UCD Conway Institute Core Facilities: Advanced Light Microscopy, Flow Cytometry, Mass Spectrometry, Genomics, and Proteomics platforms.* • *Curated chemical screening libraries and target validation datasets for ocular compounds.*

Training capacity and knowledge-transfer potential
• *Hands-on training for incoming staff in larval zebrafish handling, ocular phenotyping, visual behavioral assays, and high-content drug screening.* • *Transfer of skills in ocular pharmacology, experimental design for small molecule discovery, bioimaging techniques, and multi-omics data integration.*

What your team would hope to gain from participating
• *Access to advanced cell/tissue engineering models, human retinal organoid platforms, and novel biomaterials/drug delivery formulations.* • *Clinical exposure, translational validation pathways, and collaboration with non-academic/SME partners for downstream commercialization and regulatory navigation.*

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
tbc					

Main disciplines represented in your team	Ocular pharmacology, visual neuroscience, drug discovery, developmental biology, genetics, chemical biology
Experience in EU / international / multi-partner collaborative projects	Extensive experience coordinating and participating in EU FP7, Horizon 2020, Horizon Europe actions, and Marie Skłodowska-Curie Networks (ITN/DN, RISE/SE).
Experience with clinical, translational or patient-facing research (if relevant)	☑ Yes : Translational screening pipelines focused on identifying leads or biomarkers for clinical-stage development and commercial spin-outs.

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

1: Ruddin G, McCann T, Kaylor JJ, Fox MM, Fehilly JD, Ward R, Faulkner A, Moran AL, Wynne K, Radu RA, Monaghan MG, Thorpe SD, Travis GH, Kennedy BN. Dual Knockout Models of the Spatially and Functionally Conserved rgra and rgrb Zebrafish Genes Reveal the Requirement of RGR for the Integrity of Cone-Mediated Photopic Vision, the Photopic Visual Cycle and Bruch's Membrane Morphology. FASEB J. 2026 May 31;40(10):e71877. doi: 10.1096/fj.202504935R. PMID: 42160018.

2: Fehilly JD, McCann T, Ruddin G, Kaylor JJ, Grenville H, Ward R, Wynne K, Sánchez AG, Strachan E, Travis GH, Collery RF, Kennedy BN. Germline Disruption of Retinal Pigment Epithelium-Expressed Zebrafish rlbp1b^{-/-} Results in Selective Dim Light Visual Behavior Deficits and Provides a Screening Platform for Evaluating the Pathogenicity of Human RLBP1 Variants. FASEB J. 2025 Jun 30;39(12):e70754. doi: 10.1096/fj.202500600R. PMID: 40552921; PMCID: PMC12186719.

3: McCann T, Sundaramurthi H, Walsh C, Virdi S, Alvarez Y, Sapetto-Rebow B, Collery RF, Carter SP, Moran A, Mulholland R, O'Connor JJ, Taylor MR, Rauch N, Starostik MR, English MA, Swaroop A, Geisler R, Reynolds AL, Kennedy BN. Emc1 is essential for vision and zebrafish photoreceptor outer segment morphogenesis. FASEB J. 2024 Oct 15;38(19):e70086. doi: 10.1096/fj.202401977R. PMID: 39360639.

4: Tonelotto V, Costa-Garcia M, O'Reilly E, Smith KF, Slater K, Dillon ET, Pendino M, Higgins C, Sist P, Bosch R, Passamonti S, Piulats JM, Villanueva A, Tramer F, Vanella L, Carey M, Kennedy BN. 1,4-dihydroxy quininib activates ferroptosis pathways in metastatic uveal melanoma and reveals a novel prognostic biomarker signature. Cell Death Discov. 2024 Feb 10;10(1):70. doi: 10.1038/s41420-023-01773-8. PMID: 38341410; PMCID: PMC10858877.

5: Gómez Sánchez A, Colucci P, Moran A, Moya López A, Colligris B, Álvarez Y, Kennedy BN. Systemic treatment with cigarette smoke extract affects zebrafish visual behaviour, intraocular vasculature morphology and outer segment phagocytosis. Open Res Eur. 2023 Nov 23;3:48. doi: 10.12688/openreseurope.15491.2. PMID: 38283058; PMCID: PMC10822043.

Up to 5 relevant previous projects, grants, networks or collaborations

EVEREST (Horizon Europe MSCA Staff Exchanges)
Role: Project Coordinator (Led by UCD)
Focus: Harnessing extracellular vesicles (EVs) as non-invasive diagnostics and targeted, personalized therapies for complex conditions such as cancer and cardiovascular diseases.
Consortium Composition: 21 academic and industry institutions across 11 countries (including Trinity College Dublin, SiriusXT, Luxembourg Institute of Science and Technology, and BioReperia).

CRYSTAL3 (Horizon 2020 MSCA RISE)
Role: Consortium Coordinator
Focus: Exploring novel drug discovery and therapeutics focused on inflammation-related signaling and eye/cancer diseases through an industry-academia exchange framework.

3D-NEO-NET (Horizon 2020 MSCA RISE)
Role: Consortium Coordinator
Focus: Developing 3D disease models and identifying novel therapeutics for ocular diseases and ocular oncology (particularly uveal melanoma) by bridging academic research with biotechnology enterprise partners.

3D-NET (FP7 Industry-Academia Partnerships and Pathways - IAPP)
Role: Consortium Coordinator

Focus: Drug discovery and development targeting ocular diseases, integrating zebrafish functional genomics, targeted drug screening, and delivery solutions across academic and corporate partners.

CURAM – MedTrain+ (Horizon Europe MSCA Fellowship Program)
Role: Participating Principal Investigator / Supervisor
Focus: Identification, characterization, and optimization of novel immune-engineered peptides to treat metastatic uveal melanoma.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
☐ ***WO2014012889A1*** *— Anti-angiogenic compounds* ☐ ***US9388138B2 / US8916586B2*** *— Anti-angiogenic compounds* ☐ ***EP2875008A1*** *— Anti-angiogenic compounds* ☐ ***US20150191428A1*** *— Anti-angiogenic compounds*

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	☐ Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only ☐ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
1. *Rapid, cost-effective in vivo small-molecule screening and target validation in zebrafish models of retinal degeneration and ocular vascular disease.* 2. *Automated visual function and behavioral assays (OKR) for quantitative functional recovery testing.* 3. *Comprehensive core infrastructure for high-resolution ocular imaging, proteomics, and multi-omics analysis.*

Capabilities or expertise you would especially like to access through the consortium
Human cell/tissue platforms, retinal organoid models, bio-formulation/nanomedicine delivery, and clinical validation pathways.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	☐ Yes ☐ No
☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☐ **Understood**

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Date completed: 18/09/2026

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